CompTIA Network+ (N10-009)

Chapter 7: Given a Scenario, Use Appropriate IPv4 Network Addressing

A Study Guide for Aspiring Network Professionals

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Introduction

Understanding IPv4 addressing is foundational to every networking role. Whether you are configuring a small office LAN or designing an enterprise-level infrastructure, you will constantly rely on IP addressing concepts. Chapter 7 of the CompTIA Network+ N10-009 curriculum covers everything from binary math to advanced subnetting — topics that are heavily tested on the exam and essential in day-to-day IT work.

This guide walks through every exam objective in plain language, with real-world context, tables, and worked examples to help you build genuine understanding rather than mere memorization.

1. Binary

Everything in computing — including IP addresses — is represented in binary (base-2). Binary uses only two digits: 0 and 1. Each binary digit is called a ***bit***. Eight bits together form one byte, also called an octet in IP addressing.

Why Binary Matters for IP Addressing

IPv4 addresses are ***32 bits*** long, divided into ***four octets***. When your router applies a subnet mask to an IP address, it is performing binary AND operations at the hardware level. Mastering binary lets you subnet quickly, troubleshoot misconfigured masks, and understand how devices determine whether a destination is local or remote.

Decimal ↔ Binary Conversion

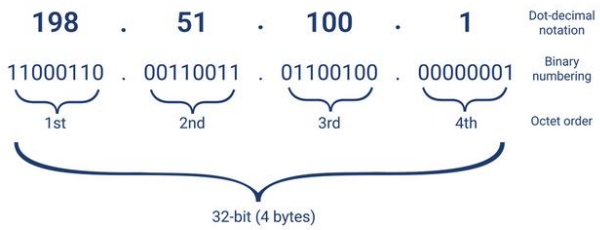
Each bit position in an octet represents a ***power of 2***, from right to left: 128, 64, 32, 16, 8, 4, 2, 1. To convert a decimal number to binary, subtract the largest power of 2 that fits, place a 1 in that position, and repeat.

Bit Position	7	6	5	4	3	2	1	0
Value (2^n)	128	64	32	16	8	4	2	1
Example: 192	1	1	0	0	0	0	0	0
Example: 168	1	0	1	0	1	0	0	0
Example: 255	1	1	1	1	1	1	1	1

Exam Tip: *Practice converting common octets: 192, 172, 10, 255, 128, 0. You will encounter them repeatedly in subnetting questions.*

2. IP Addressing

An IPv4 address is a 32-bit number written in dotted-decimal notation — ***four octets*** separated by dots, such as 192.51.100.1. Every device on a TCP/IP network requires a unique IP address to communicate. The address has two logical parts: the ***network portion*** (identifies the network) and the ***host portion*** (identifies the specific device).



This figure shows the breakdown of the IP address 198.51.100.1. Each octet has its own numbering-for instance, 198 is transmitted as 11000110

There are some large blocks that are reserved for special purposes to ensure propre functioning, management, and efficiency of the internet. These are as follows:

- 100.64.0.0/10: Contains-grade NAT, used by ***ISPs*** to manage large-scale address translation
- 192.0.0.0/24: Reserved for IETF protocol assignments
- 192.88.99.0/24: Used for 6to4 relay anycast, helping IPv6 and IPv4 coexist
- 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24: Set aside for ***documentation*** and ***example purposes*** in educational materials

These reserved IP ranges help avoid conflicts in real-world network operations and ensure that certain tasks, such as testing example creation, and protocol development, are handled properly without affecting live internet traffic.

Public vs. Private Addressing

IP addresses are divided into two broad categories: public and private. Public addresses are globally routable on the Internet. Private addresses (***RFC 1918***) are reserved for internal networks and are not routed on the Internet — they require Network Address Translation (NAT) to communicate outside the LAN.

Range	Type	Class	Typical Use
10.0.0.0 – 10.255.255.255	Private	A	Large enterprises / ISPs
172.16.0.0 – 172.31.255.255	Private	B	Medium-sized networks
192.168.0.0 – 192.168.255.255	Private	C	Home / SOHO networks
All other addresses	Public	A/B/C	Internet-routable devices

Automatic Private IP Addressing (APIPA)

APIPA is a **self-configuration** mechanism built into Windows (and supported by other OSes). When a device cannot reach a DHCP server, it automatically assigns itself an address in the range 169.254.0.0 – 169.254.255.255 (with subnet mask 255.255.0.0). APIPA enables link-local communication on the same subnet, but the device will not have a default gateway, so it cannot reach remote networks or the Internet.

- APIPA range: 169.254.0.0 – 169.254.255.255 (/16)
- Assigned automatically when DHCP fails
- Allows local subnet communication only — no Internet access
- Seeing a 169.254.x.x address on a client is a classic sign of DHCP failure

> **Exam Tip:** *If a user complains they cannot access the Internet and their IP starts with 169.254, suspect a DHCP server issue first.*

Loopback / Localhost

The loopback address is used to **test the TCP/IP stack** on a local device without sending traffic to the network. The most commonly used loopback address is 127.0.0.1, often called ***localhost***. The entire range 127.0.0.0/8 is reserved for loopback, but 127.0.0.1 is the conventional address.

- Loopback range: 127.0.0.0 – 127.255.255.255
- Most common: 127.0.0.1 (localhost)
- Pinging 127.0.0.1 confirms the TCP/IP stack is functioning locally
- Traffic sent to loopback never leaves the host machine

```
C:\Users\User>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. IP Address Classes

The original IPv4 design used ***classful*** addressing, dividing the address space into ***five classes*** (A through E) based on the value of the first octet. Although modern networking uses ***classless addressing*** (***CIDR***), understanding IP classes remains important for the Network+ exam and for interpreting legacy documentation.

Class	First Octet Range	Default Subnet Mask	Network Bits	Host Bits	Max Hosts/Network
A	1 – 126	255.0.0.0 (/8)	8	24	16,777,214
B	128 – 191	255.255.0.0 (/16)	16	16	65,534
C	192 – 223	255.255.255.0 (/24)	24	8	254
D	224 – 239	N/A (Multicast)	N/A	N/A	N/A
E	240 – 255	N/A (Experimental)	N/A	N/A	N/A

Hosts

Hosts are individual devices on a network that have an IP address — computers, printers, phones, servers, etc. The number of usable host addresses in a subnet is calculated as **$2^n - 2$**, where n is the number of host bits. We subtract 2 because the first address (***network address***) and last address (***broadcast address***) cannot be assigned to devices.

- /24 subnet (8 host bits): $2^8 - 2 = 254$ usable host addresses
- /16 subnet (16 host bits): $2^{16} - 2 = 65{,}534$ usable host addresses
- /8 subnet (24 host bits): $2^{24} - 2 = 16{,}777{,}214$ usable host addresses

Networks

The network address is the first IP address in a subnet — all host bits are set to 0. It identifies the subnet itself and cannot be assigned to any device. For example, in the subnet 192.168.1.0/24, the network address is 192.168.1.0 and the broadcast address is 192.168.1.255.

Class A: Uses 7 bits because the first bit is always set to 0 to indicate a **Class A** address.

2^8=128 networks. You cannot use 127.x.x.x as it is a loopback address, and 0.x.x.x is reserved for special usage.

Class B: Uses 14 bits because the first 2 bits are set to 10 to indicate a **Class B** address.

2^{14}=16,384 networks.

Class C: Uses 21 bits because the first 3 bits is always set to 110 to indicate a **Class C** address.

2^{21}=2,097,152 networks.

Subnet Masks

A subnet mask is a 32-bit number that tells devices which portion of an IP address is the network and which is the host. Network bits are represented as 1s and host bits as 0s. Masks are written either in dotted-decimal (255.255.255.0) or in CIDR prefix notation (/24).

CIDR Prefix	Dotted-Decimal	Binary Representation	# of Host Bits	Usable Hosts
/8	255.0.0.0	11111111.00000000.00000000.00000000	24	16,777,214
/16	255.255.0.0	11111111.11111111.00000000.00000000	16	65,534
/24	255.255.255.0	11111111.11111111.11111111.00000000	8	254
/25	255.255.255.128	11111111.11111111.11111111.10000000	7	126
/26	255.255.255.192	11111111.11111111.11111111.11000000	6	62
/27	255.255.255.224	11111111.11111111.11111111.11100000	5	30
/28	255.255.255.240	11111111.11111111.11111111.11110000	4	14
/29	255.255.255.248	11111111.11111111.11111111.11111000	3	6
/30	255.255.255.252	11111111.11111111.11111111.11111100	2	2

4. Subnetting

Subnetting is the process of ***dividing a single***, large network into smaller, more manageable sub-networks (subnets). This is done by ***borrowing bits*** from the ***host portion*** of the address and using them to create additional network identifiers.

Advantages of Subnetting

- ***Reduced broadcast traffic*** — broadcasts are contained within each subnet, improving performance
- ***Improved security*** — subnets can be isolated by firewalls and ACLs, limiting attack spread
- ***Efficient IP address utilization*** — allocate address space proportional to actual need
- ***Simplified management*** — smaller subnets are easier to monitor and troubleshoot
- ***Hierarchical routing*** — routers can summarize routes, reducing routing table size
- ***Organizational alignment*** — subnets can map to departments, buildings, or VLANs

Devising a Subnetting Plan

Before subnetting, answer three questions: How many subnets do you need? How many hosts per subnet? What is your address block? Use this formula to determine how many bits to borrow:

- Subnets needed → borrow n bits where $2^n \geq$ required subnets
- Hosts needed → keep m bits where $2^m - 2 \geq$ required hosts
- Verify: $n + m \leq$ remaining host bits in original class

Worked Example: *You need 6 subnets from 192.168.10.0/24 with at least 20 hosts each. Borrow 3 bits → 2^3=8 subnets ✓. Remaining host bits: 8−3=5 → 2^5−2=30 hosts ✓. New mask: /27 (255.255.255.224). Subnets: .0, .32, .64, .96, .128, .160, .192, .224*

Subnet Calculation Quick Reference

Bits Borrowed	New Subnets	New Prefix (from /24)	Hosts/Subnet	Block Size
1	2	/25	126	128
2	4	/26	62	64
3	8	/27	30	32
4	16	/28	14	16
5	32	/29	6	8
6	64	/30	2	4

Working out Subnets with the Lone Ranger table

If you do not have any experience in subnetting, you can use a simple method to work out the masks and networks, as shown in the Table

Bits	1	2	3	4	5	6	7	8
Range (2^n)	128	64	32	16	8	4	2	1
Subnet Mask	128	192	224	240	248	252	254	255
Networks	2	4	8	16	32	64	128	256

Subnetting example

Your IT department has purchased the ***Class B*** IP address ***131.107.0.0***. Because it is Class B, it has the default subnet ***255.255.0.0***. This means the ***first two octets*** are the ***network ID*** and the second two octets are for hosts. The business requires ***8 subnets***. We use the Lane Ranger table to look for 8 subnets. From this, we can see that we need to ***borrow 3 bits*** from the host to create more networks. For the network/subnet ID, that means we will be customizing that third octet of the IP address. We can see from the lookup table that for 3 bits, the ***subnet mask*** is ***224***.

Bits	1	2	3	4	5	6	7	8
Range	128	64	32	16	8	4	2	1
Subnet Mask	128	192	224	240	248	252	254	255
Networks	2	4	8	16	32	64	128	256

This means that octet 3 for the subnet mask should be 224. So, the subnet mask is now ***255.255.224.0.*** Because we have borrowed 3 bits from octet 3, there are 5 bits remaining, in addition to the last octet. The ***range*** is ***32***

The next thing to do is to work out the address ranges for each of the subnets.

- The address that we purchased was ***131.107.0.0,*** so the values in the ***first 2 octets will not*** change***.***
- The start value for ***octet 3*** is ***0***, with the subsequent values ***increasing*** by ***32*** with each subnet - so 32, then 64, then 96, and so on.
- The end value for ***octet 3*** is ***31***, with the subsequent values ***increasing*** by ***32*** with each subnet - so 63, then 95, then 127, and so on.
- We know that an IP address for a host ***cannot start*** at ***0***; therefore, for each start IP address, the host portion is ***1***.
- We also know that IP address ***cannot end*** in ***255***; therefore, for each end IP address, the host portion is ***254***

	Start IP Address					End IP Address			
Subnet 1	131	107	0	1		131	107	31	254
Subnet 2	131	107	32	1		131	107	63	254
Subnet 3	131	107	64	1		131	107	95	254
Subnet 4	131	107	96	1		131	107	127	254
Subnet 5	131	107	128	1		131	107	159	254
Subnet 6	131	107	160	1		131	107	191	254
Subnet 7	131	107	192	1		131	107	223	254
Subnet 8	131	107	224	1		131	107	255	254

Magic Number

The ***magic number*** is a tool that will help you understand whether computers are in the same subnet without the need for any special tools- just a piece of paper. In the following question need to find out which of these machines can work on the same subnet.

	IP Address	Subnet Mask
Server 1	158.17.31.254	255.255.224.0
Server 2	158.17.32.1	255.255.224.0
Server 3	158.17.60.1	255.255.224.0

- ***Step1***: Ensure that all the servers have the same subnet mask, which they do here.
- ***Step2***: Going from left to right, look at the first value that is ***not a zero***. You can see this is *224*.
- ***Step3***: Subtract 224 from the magic number, which is always ***256.*** 256-224=32, this gives us the range – in this case 32
- ***Step4***: write out the first 4 subnets or until you cover all the numbers. Remember, the first subnet starts at 0, then we add 32.

Because the mask is the third octet, we only need the third octet from the IP addresses. Server 1 is 56, server 2 is 32, server 3 is 30. Place those values against the relevant subnet.

Subnet 1	0	0 to 31	Server 3
Subnet 2	32	32 to 63	Server 1, Server 2
Subnet 3	64	64 to 95	
Subnet 4	96	96 to 127	

This is a useful skill for real-world networking, as well as in the CompTIA exam.

Classless Inter-Domain Routing (CIDR)

CIDR (pronounced 'cider') was introduced in 1993 to replace classful addressing and slow the exhaustion of IPv4 addresses. CIDR allows network prefixes of any length, not just the fixed /8, /16, or /24 boundaries of classful addressing.

CIDR notation appends a forward slash and the prefix length to the IP address (e.g., 10.0.0.0/22). This single notation replaces the need to separately state the address and subnet mask.

- Eliminates class boundaries — any prefix length from /1 to /32 is valid
- Route aggregation (supernetting) — ISPs combine multiple prefixes into one advertisement
- Reduces routing table size — key to Internet scalability
- Enables VLSM (variable length subnet masking) — see next section

> **Example:** *192.168.4.0/22 covers 192.168.4.0 – 192.168.7.255 (1,022 usable hosts). In classful addressing, this would require four /24 Class C blocks.*

There are some commune masks, as shown in the following table

CIDR Notation	Dotted Decimal Notation	Number of Host Addresses
/8	255.0.0.0	16,777,216
/16	255.255.0.0	65,534
/24	255.255.255.0	254
/30	255.255.255.252	0

Variable Length Subnet Mask (VLSM)

VLSM allows different subnets within the same network to use different prefix lengths — meaning each subnet can be sized exactly to fit the number of hosts it needs. This maximizes address efficiency and is standard practice in modern network design.

How VLSM Works

Allocate the largest subnets first, then carve smaller subnets from remaining space:

- Step 1 — List all required subnets ordered from largest to smallest host count
- Step 2 — Assign the smallest mask that satisfies the largest requirement
- Step 3 — Take the next available block for the second subnet, sized appropriately
- Step 4 — Repeat until all subnets are assigned
- Step 5 — Verify no subnets overlap

VLSM Example: *Given 10.0.0.0/24: Dept A needs 60 hosts → /26 (10.0.0.0–10.0.0.63). Dept B needs 25 hosts → /27 (10.0.0.64–10.0.0.95). WAN link needs 2 hosts → /30 (10.0.0.96–10.0.0.99). All fit within /24 with no overlap.*

VLSM vs. Fixed-Length Subnetting

Feature	Fixed-Length (FLSM)	Variable-Length (VLSM)
Subnet sizes	All equal	Different per subnet
Address efficiency	Lower (wasted space)	Higher (right-sized)
Complexity	Simpler	More complex
Routing protocol support	All protocols	Requires classless protocols (OSPF, EIGRP, BGP)
Use case	Small, uniform networks	Enterprise / mixed-size networks

5. Quick Reference: Key Formulas & Facts

Concept	Formula / Value
Usable hosts per subnet	$2^n - 2$ (n = host bits)
Number of subnets created	2^s (s = bits borrowed)
Block size (subnet increment)	256 – decimal value of last non-zero octet
Loopback range	127.0.0.0 – 127.255.255.255
APIPA range	169.254.0.0 – 169.254.255.255
Private Class A range	10.0.0.0 – 10.255.255.255
Private Class B range	172.16.0.0 – 172.31.255.255
Private Class C range	192.168.0.0 – 192.168.255.255
Default Class A mask	255.0.0.0 (/8)
Default Class B mask	255.255.0.0 (/16)
Default Class C mask	255.255.255.0 (/24)

6. Exam Tips & Common Pitfalls

Things to Memorize

- Private IP ranges (RFC 1918) — you will see them in almost every subnetting question
- Powers of 2 up to 2^{16}: 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536
- Common subnet masks and their CIDR equivalents (especially /24 through /30)
- APIPA = 169.254.x.x means DHCP failed
- 127.0.0.1 = loopback; pinging it tests the local TCP/IP stack

Common Pitfalls

- Forgetting to subtract 2 for network and broadcast addresses when counting usable hosts
- Confusing the number of subnets with the number of hosts
- Treating 127.x.x.x as a usable address range — it is reserved for loopback
- Overlooking that Class D (224–239) is multicast and Class E (240–255) is experimental
- Mixing up VLSM and CIDR — CIDR is a notation system; VLSM is a design technique enabled by CIDR

Conclusion

IPv4 addressing is not just an exam topic — it is the backbone of virtually every enterprise network you will work with in your career. By mastering binary conversion, understanding the distinction between public and private addressing, and confidently applying subnetting and VLSM, you will be well-equipped both for the Network+ exam and for real-world networking challenges.

Practice subnetting daily. Use tools like subnet calculators to check your work, but aim to perform calculations by hand before the exam. Consistent practice will make these calculations second nature.

> **Next Steps:** *Review the practice questions in your study guide, take a timed subnetting quiz, and lab the concepts using Packet Tracer or GNS3 to reinforce hands-on understanding.*